

Project Goal

Goal: Design and develop a device that aids clinicians in monitoring symptoms and recovery trends in adolescent and young adult concussion patients.

Parameters: Symptom tracking, cognitive and sensory feedback, balance and reaction time data, longitudinal trend analysis.

Background

Current challenges

- Most tools rely on subjective self-reporting and visual examination (e.g., SCAT5 Sports Concussion Assessment Tool, 5th Edition)^[1]
- Current concussion symptom tracking involves verbal symptom checklists and clinically-administered cognitive tests.
- Research shows grip strength, balance, and reaction time are valid recovery metrics^{[2][3]}



SCAT5 Testing Procedure



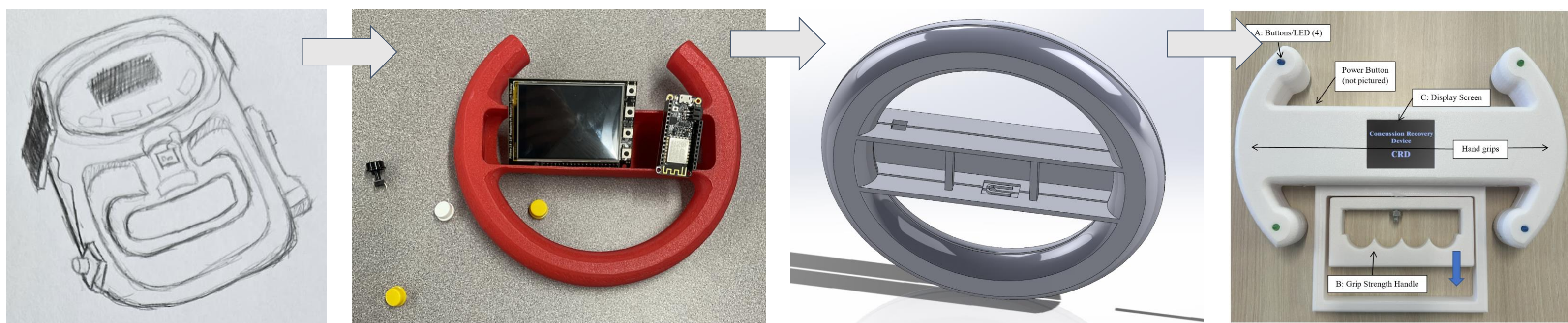
SCAT5 On Site Testing

Our Solution

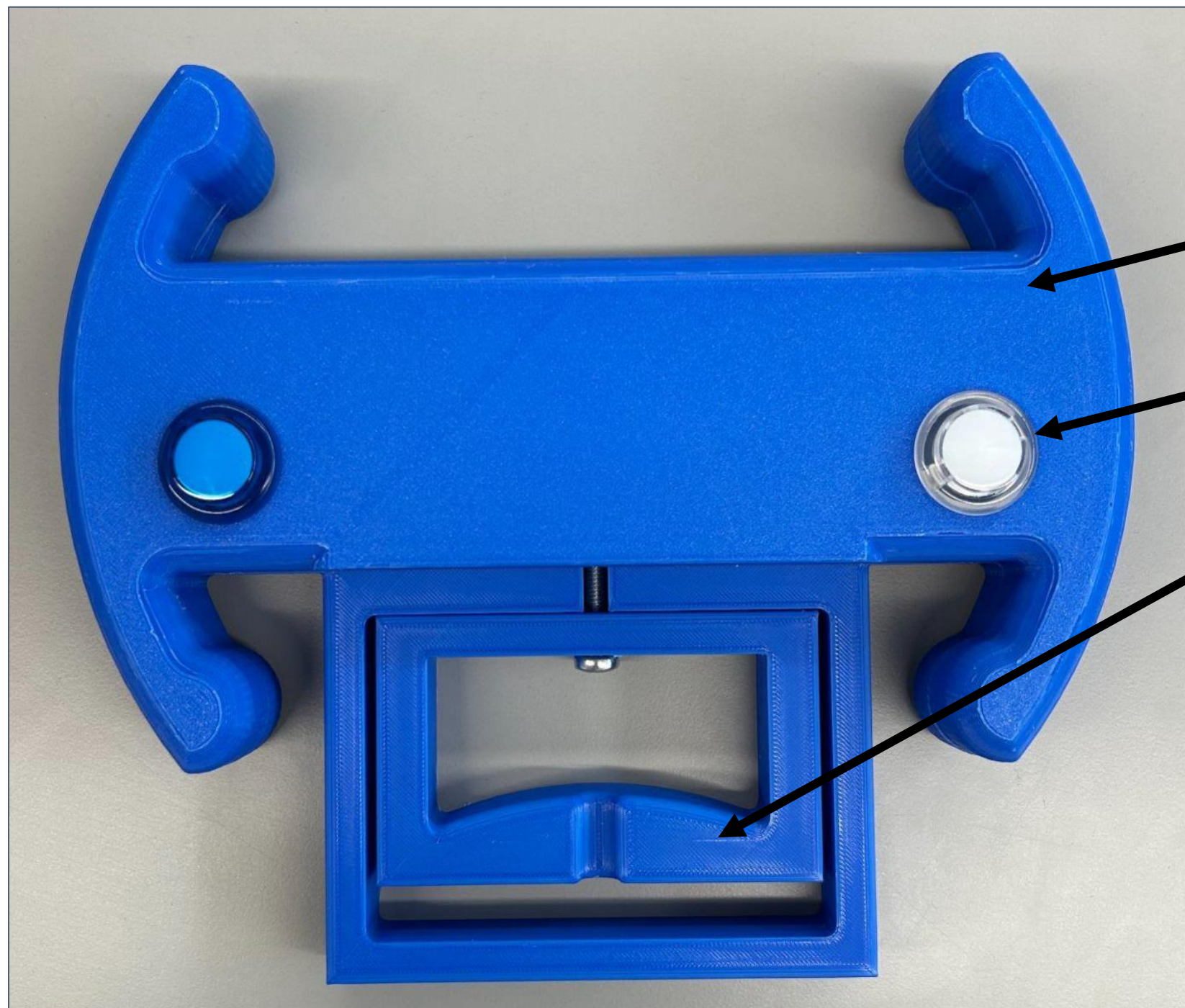
- Our device objectively measurements of reaction time, grip strength, and balance.
- Thresholds are based on average values by gender (critical for grip strength), orientation of center of mass (balance), and average expected result (reaction time).
- If a user exceeds these thresholds, the device measures deviation from standard.

Design Process

Design Iteration



Final Design



Externals

Body: 3D printed with PLA

LED buttons for interaction and reaction time

Handle for grip strength testing

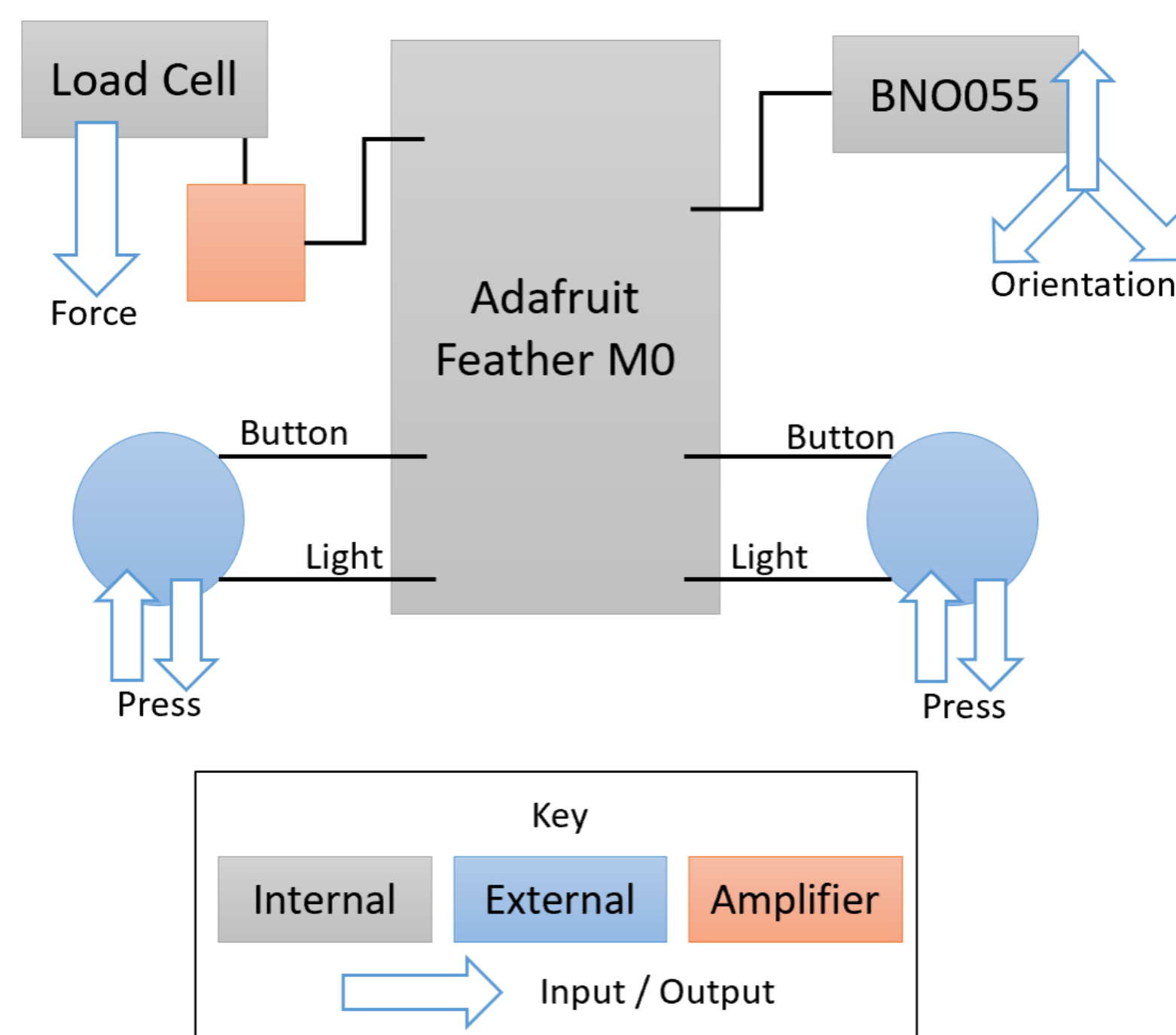
Internals

Adafruit Feather M0 microprocessor

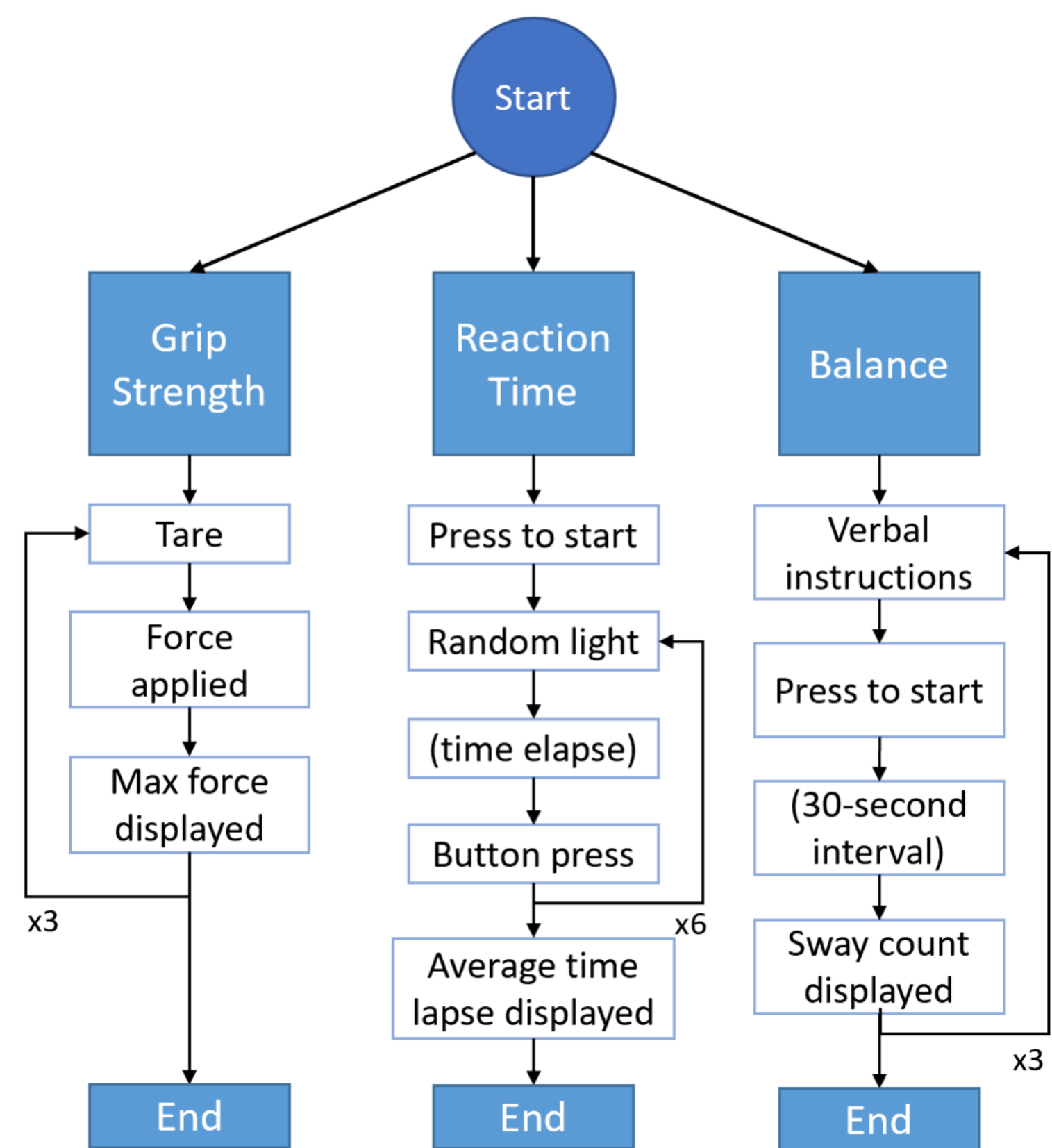
Adafruit BN055 orientation sensor

Load cell and amplifier (0-200 lbs)

Circuit Diagram



Programming Functions



User Feedback

11 subjects (average)

Comfort



Accuracy



Useability

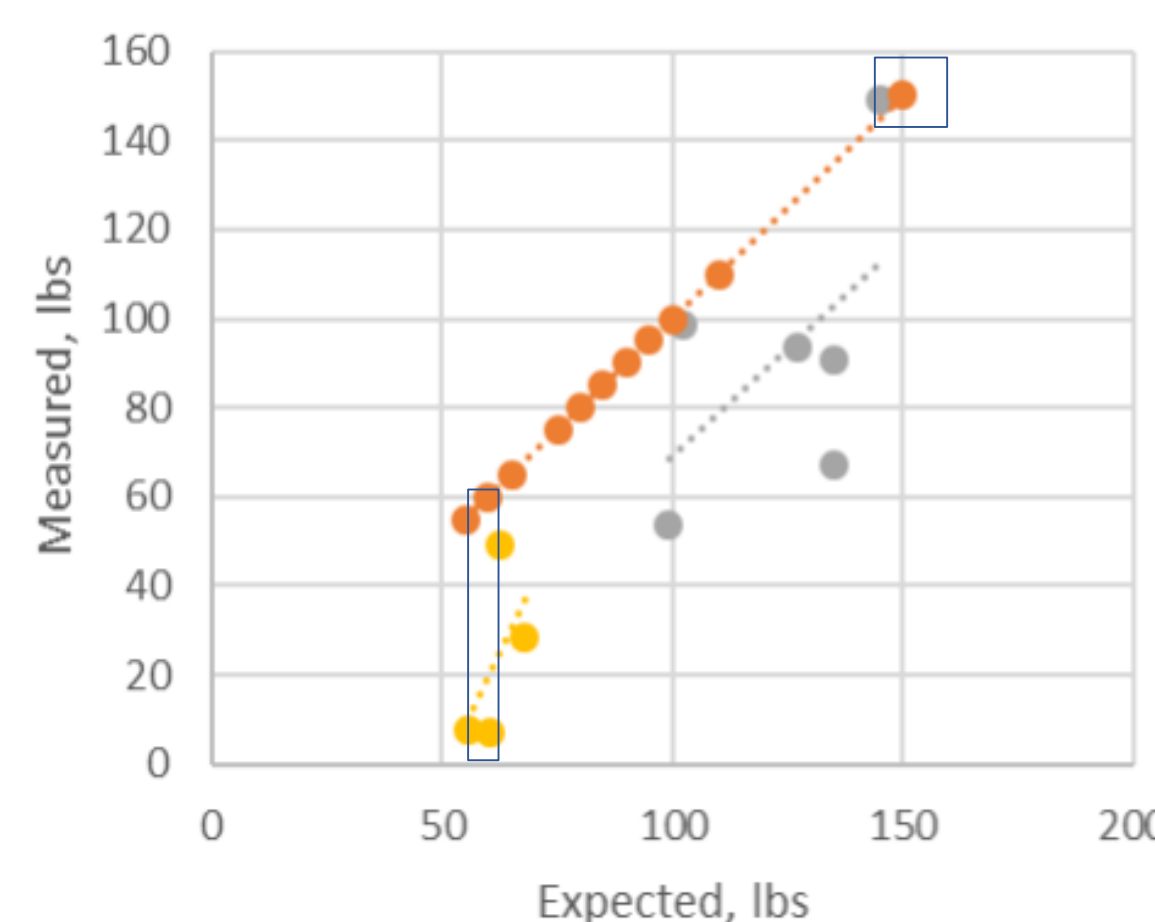


Integration



Results and Conclusions

Grip Strength



Testing Method

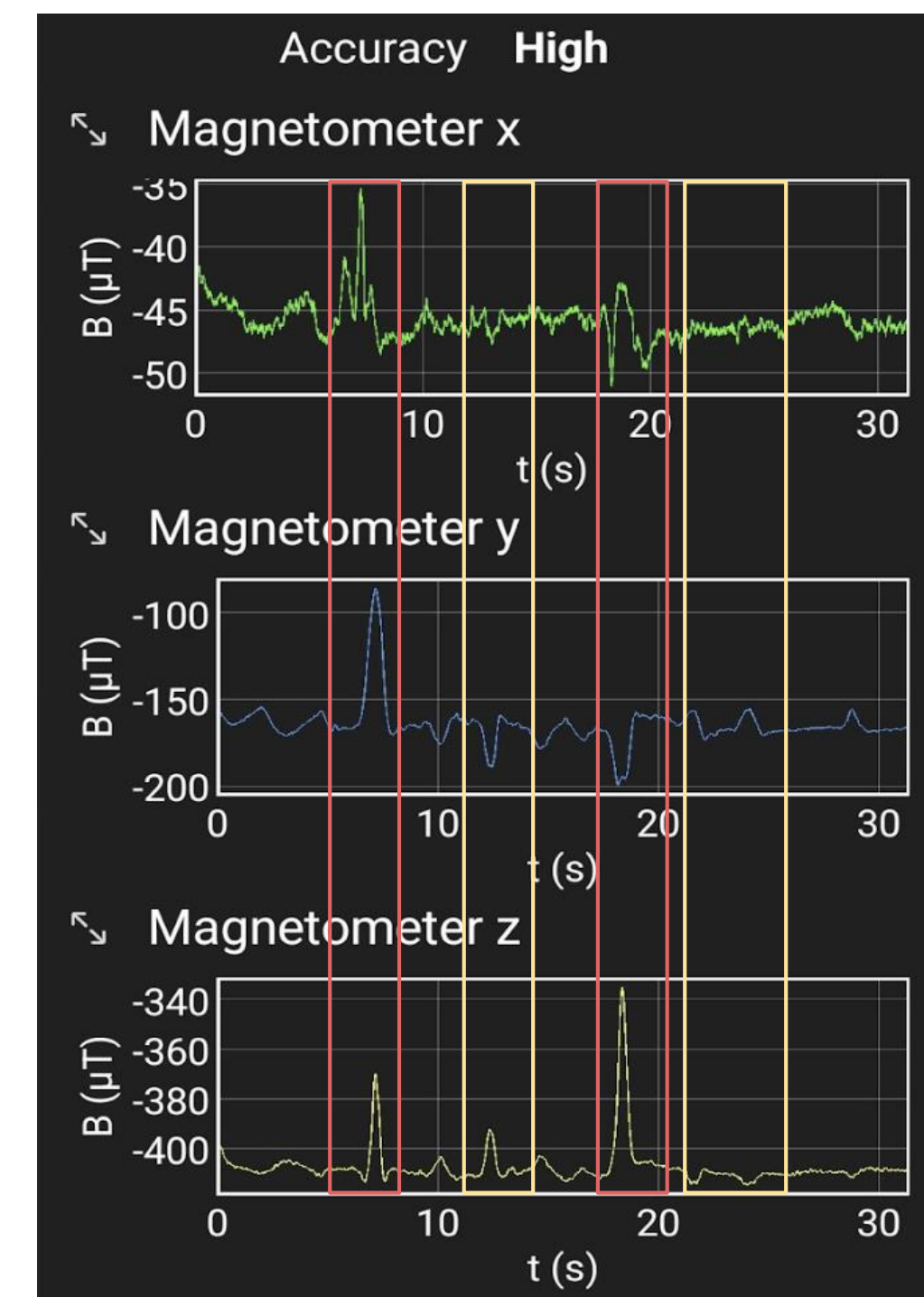
11 participants tested (6 male, 5 female)

Calibrated using known hand weights, with expected measurements taken by hand grip dynamometer.

Minimum percent error: 2.9% (Male)

Maximum percent error: 88% (Female)

Balance



Representative Test

Stable for 26/30 sec

Moderate: 2

Severe: 2

Moderate Sway

Severe Sway

Testing Method

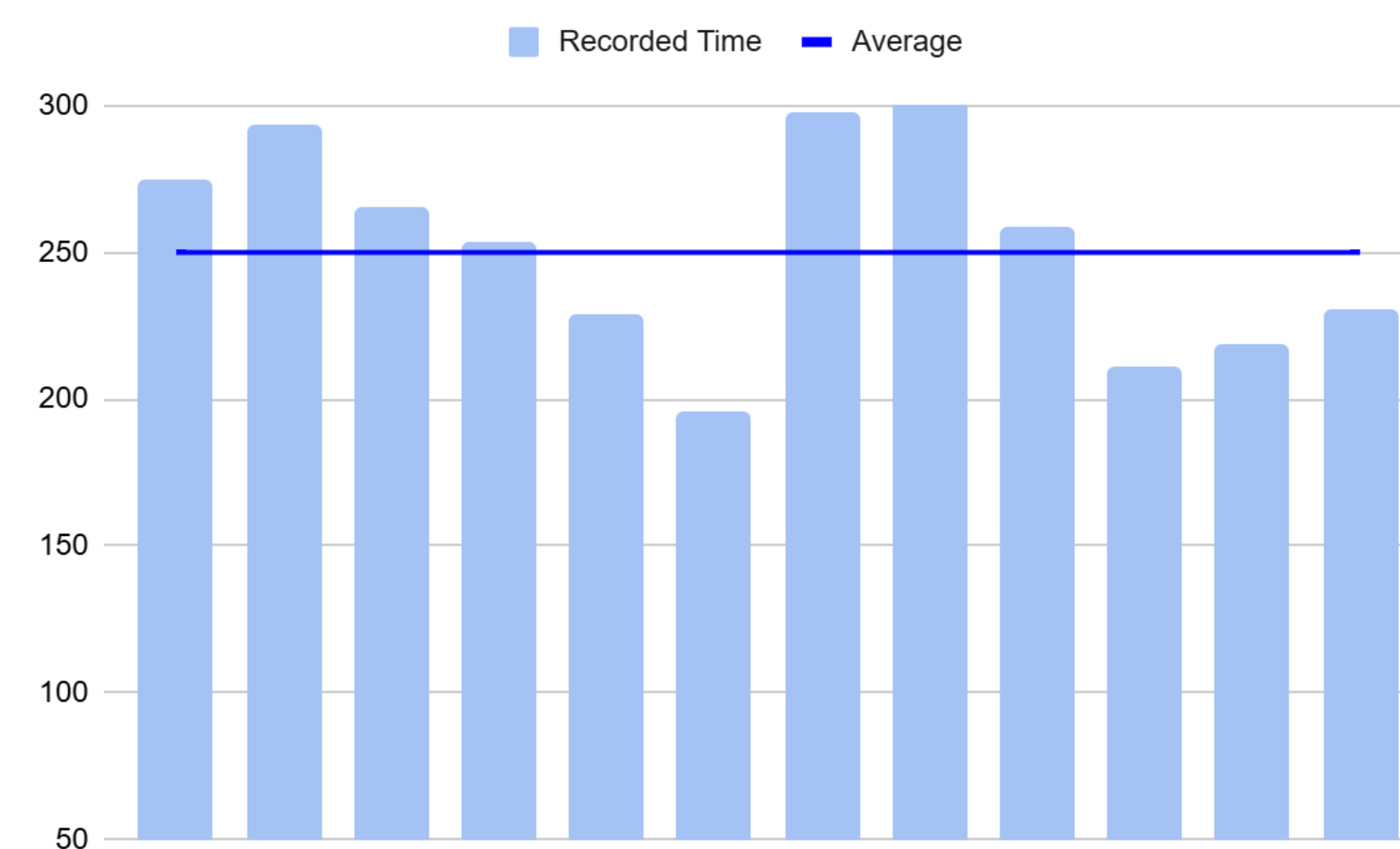
Orientation data collected with both Phyphox magnetometer (orientation) data simultaneously. Reported sways correlated to spikes in orientation shift.

Tolerances

Moderate sway: Euler angle > 15°

Severe sway: Euler angle > 30°

Reaction Time



Testing Method

12 participants (7 female, 5 male)

6 reaction time events per test, with the average time plotted against a standard average reaction time.

Fastest time recorded: 195 ms

Slowest time recorded: 353 ms

- Grip strength sensor used to capture muscular function related to concussion recovery, specialized to gender.
- Balance testing using integrated sensor system provided real-time sway detection and improved accuracy and consistency of physical performance metrics.
- Reaction time measurements offered quantifiable cognitive feedback beyond subjective scoring.

Future Work

User Interface Enhancements

- Adding an LED screen for enhanced user interaction and intuitive control.
- Store user data safely to show time history trends across multiple users to better track recovery and simplify clinician review.
- Better ergonomics and adjustability in grip strength handle for more accurate data collection and user comfort.

Electrical Enhancements

- Increase sensor accuracy with higher-quality components.
- Design a custom PCB to improve system reliability.
- Adding a rechargeable battery to create a stand-alone device for sideline testing.

Future Capabilities

- Adding more sensors for more well-rounded symptom quantification after a concussion.
- Bluetooth data transmission for integration into current patient data systems.

Acknowledgements and References

We would like to thank Professor John Peponis, Andrew Ulaszek and the rest of the Biomedical Engineering Department faculty for their support on this project.

[1] Koren, L. (2024) *Concussion Assessment Tools: When and How to Use Them*.

[2] Reed, N. (2016) *A Preliminary Exploration of Concussion and Strength Performance in Youth Ice Hockey Players*.

[3] McLeod, T. (2014) *Vestibular and Balance Issues Following Sport-related Concussion*.